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A structured approach for enhancing clinical risk monitoring and workflow digitalization in healthcare

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ABSTRACT

A multitude of studies have highlighted alarming data on healthcare adverse events and medical errors in many countries, including Italy. Many of these errors appear to be the result of non-compliance with good practices and organizational operating procedures, specified in ministerial directives and international standards. This paper describes the efforts of an Italian research project to propose a structured approach to healthcare process management, emphasizing clinical risk management. Using the Business Process Model and Standard Notation (BPMN), the project models clinical processes and integrates them into the clinical information system to identify process deficiencies, highlight compliance vulnerabilities, monitoring and managing clinical risks. The workflows' digitalization, enabled the definition and calculation of several Key Performance Indicators (KPIs) for a long-term evaluation of the success of the modeled processes. In the experimental phase, the project allowed to identify areas significantly affected by operational deviations, enabling targeted actions to safeguard patient health and generate economic savings due to the reduction of legal actions against medical workers' errors. The effects of an overall improvement on quality of care can be appreciated not only by patients but also by medical and administrative staff of the clinics.

1. Introduction

The rapid progress in technology and the integration of information systems in complex business and organizational settings have prompted the adoption of well-structured, secure, and standardized methodologies for contemporary production processes. These advancements have not only enhanced efficiency and control but have also enabled organizations to adapt swiftly to evolving demands in the global market [1]. Customers now seek high-quality products delivered in shorter time frames and customized to their specific needs. Therefore, any deviation from established compliance standards can result in substantial economic and reputations costs. It has become crucial for businesses to maintain stringent adherence to these standards to ensure seamless operations and customer satisfaction in the competitive market landscape. Indeed, the demand for conformity and efficiency has transcended the industrial sector and has become equally important in the services domain [2–5]. This transformation is largely driven by the widespread integration of Internet of Things (IoT) technologies, cloud computing and Artificial intelligence (AI). These advancements have revolutionized the way services are delivered, allowing for unprecedented levels of customization, scalability, and resilience in service offerings [6,7].

The concept of Smart Hospital aligns with this promising trend, utilizing modern technologies and Information Technology (IT) infrastructures to revolutionize healthcare practices [8,9], leading to improved accuracy, reduced errors, enhanced efficiency [10], and agility in medical procedures, particularly vital for clinical risk management [11]. Consequently, this approach indirectly contributes to reduced costs associated with compensating patients affected by medical errors or malpractice, fostering greater trust and confidence in the healthcare system [12–15].

This paper delineates the endeavors of the research project "Mo.Ri.San. Monitoring and management of clinical risk in the social and healthcare sector", aimed at proposing a structured approach to healthcare, with a particular emphasis on clinical risk management [16]. The analysis of the state of the art, reported in Section 2, highlighted that some different methodologies have been adopted to implement clinical risk management in the modern healthcare sector. Different tools and technologies are proposed in previous works for process monitoring purposes, both in the healthcare contexts and in industrial ones. Some researchers propose Blockchain and Internet of Things technologies, to improve security and reliability of clinical pathways. Furthermore, other studies used Business Process Management to improve the performance and the coordination between workers in a clinical environment. From the literature analysis, the lack of an integrated tool for real-time monitoring and data collection in a healthcare company emerged, which could also represent a support instrument for risk management decisions. Unlike those that use the

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blockchain, this tool has the advantage of being easy to modify in order to adapt to changes in the processes, easy to be monitored and used in other contexts just modifying the process maps but maintaining the proposed architecture. For this purpose, leveraging the Business Process Management (BPM) approach and using the Business Process Management standard Notation BPMN 2.0 [17], the project developed the modeling of structured and repeatable care pathways within healthcare facilities [18] and seamlessly integrates them into the clinical information system to identify process deficiencies, pinpoint compliance vulnerabilities, and effectively monitor and manage clinical risks. Then information technology and process management served as enablers to ensure patient safety and long-term economic savings for the healthcare context, characterized by complex operations and human interactions that are inherently challenging to control and standardize. The resulting insights and evaluations will guide the project's partners (two Italian private clinics) in implementing operational constraints within their software, ensuring compliance with medical and organizational procedures in line with current regulatory frameworks and reference standards.

The rest of the paper is organized as follows. Related works are reported in Section 2. In Section 3 the process modeling phase is described together with some details on a specific process. Section 5 reports the integration of the developed tool with the healthcare companies' Enterprise Resource Planning (ERP) software while Section 6 describes the experimental phase and discusses the obtained results. Finally, the conclusions of the work are proposed in Section 7.

2. Related works

The importance of clinical risk management in modern healthcare systems cannot be overstated, especially considering the impact it has on patient safety and the overall quality of care. Legislation in the healthcare sector often mandates the implementation of risk management practices to minimize adverse events and ensure patient well-being [19]. The research conducted by Kohn et al. [20], from the Institute of Medicine (IOM), sheds light on the definition of clinical risk, describing it as the likelihood that a patient may experience an adverse event or suffer harm or discomfort due to the treatment received during hospitalization. This definition emphasizes that clinical risk includes any unintentional harm or discomfort that may occur during the patient's stay. Inadequate management of clinical risk has serious consequences. It is a leading cause of legal actions against healthcare facilities in Organization for Economic Co-operation and Development (OECD) countries, indicating the significance of addressing risk proactively to avoid potential litigation and reputational damage. Moreover, the Institute for Healthcare Improvement (IHI) has identified clinical risk mismanagement as the third leading cause of death in the United States, emphasizing the urgent need to enhance risk management practices to prevent patient harm and fatalities. As a result, healthcare organizations are increasingly focused on implementing effective clinical risk management strategies to improve patient outcomes, enhance patient safety, and reduce adverse events [21]. The constantly evolving healthcare landscape, characterized by technological advancements, changing regulations, and organizational complexities, has necessitated a more comprehensive approach to clinical risk management [22]. Healthcare facilities are now focusing on proactive and systematic analysis to identify potential causes of adverse events and take preventive measures before they occur. By adopting a proactive approach to clinical risk management, healthcare facilities can create a safer environment for patients, staff, and all stakeholders involved in the healthcare process. This involves conducting in-depth analyses of potential risks and vulnerabilities, using data-driven insights to identify patterns and trends that could lead to adverse events [23]. Through such analysis, healthcare providers can implement targeted interventions, such as process improvements, training programs, and the adoption of best practices. This ensures that potential risks are mitigated or eliminated before they can adversely impact patient safety or the quality of care provided. Additionally, a proactive clinical risk management approach fosters a culture of safety and continuous improvement within healthcare organizations [24]. It encourages open communication, reporting of near-misses or potential risks, and a commitment to learning from past experiences to prevent similar incidents in the future. The implementation of advanced technology solutions, such as electronic health records, clinical decision support systems, and real-time monitoring tools, has further enhanced the ability to identify and address potential risks. These technologies enable the timely capture and analysis of patient data, facilitating the early detection of anomalies or deviations from expected outcomes. The impact of modern digital technologies in healthcare pathways has been explored in [25], emphasizing the opportunities offered by the new approaches, in terms of the resiliency of performance in healthcare. The joint adoption of digital technologies and lean interventions has been investigated in the review proposed by Tlapa et al. [26] research. In this regard, Cagliano et al. [27] demonstrated the advantages deriving from the application of a structured and systemic approach in identifying risks for the patient, by the Reason theory, within health contexts characterized, by their nature, by a strong variability linked to human decision-making processes.

Wingate [28] emphasized the potential impact of computerized systems in healthcare and pharmaceutical companies. These systems play a crucial role in supporting daily operations and ensuring regulatory compliance through IT validation systems. Computerized systems have become essential for various aspects of healthcare and pharmaceutical operations. They streamline and automate essential processes, such as patient record-keeping, medication management, inventory control, and laboratory analysis. By implementing computerized systems, healthcare facilities and pharmaceutical companies can achieve higher levels of efficiency, accuracy, and data integrity. One critical aspect of these computerized systems is their role in regulatory compliance. Healthcare and pharmaceutical industries are subject to strict regulations and guidelines to ensure patient safety and product quality. IT validation systems help these companies adhere to regulatory requirements and demonstrate the reliability and accuracy of their computerized systems.

The Access Risk Knowledge (ARK) platform, as introduced by Crotti et al. [29], represents a significant advancement in clinical risk management. This platform leverages Semantic Web technologies to effectively model, integrate, and classify risk and socio-technical system analysis information from various data sources, encompassing both qualitative and quantitative data, into a cohesive risk graph. One of the notable features of the ARK platform is the development of a clinical safety management taxonomy, which serves as an annotation system for qualitative risk data. This taxonomy enables the automated analysis of qualitative risk information, streamlining the risk management process and enhancing the accuracy and efficiency of risk assessments. By integrating diverse data sources and information into a unified risk graph, the ARK platform provides healthcare facilities with a comprehensive and holistic view of potential risks and safety concerns. This comprehensive approach allows for a more thorough understanding of the complex interactions and dependencies within the healthcare system, enabling proactive risk identification and mitigation. Another data management model for proactive risk management has been proposed in the Meshkov et al. [30] work.

Furthermore, several recent studies have proposed the implementation of Business Process Management in the healthcare sector. Among these, Emanuele and Koetter [31] analyzed a case study of integration between BPM and corporate information systems within a healthcare facility, highlighting the advantages related to the support it can give to processes.

In the study conducted by Reichert [32], the focus was on the potential adoption of Process-Aware Information Systems (PAIS) in healthcare. PAIS are information systems that incorporate healthcare business processes within their framework. The research highlighted the importance of flexibility in adapting these systems to the inherent variability that characterizes healthcare pathways, encompassing all the decision-making processes involved. Healthcare pathways are complex and dynamic, involving numerous stakeholders, multiple decision points, and diverse treatment options tailored to individual patient needs. These pathways often encounter variations due to patient characteristics, medical conditions, and external factors. As a result, a one-size-fits-all approach to healthcare processes is not suitable, and there is a growing recognition of the need for flexible and adaptive systems to handle these complexities. PAIS offer a promising solution as they are designed to be process-aware, meaning they can capture, model, and execute complex workflows that represent the dynamic nature of healthcare processes. The study by Reichert et al. underscores the importance of adopting PAIS in healthcare to meet the ever-changing demands of patient care. The adoption of such flexible and adaptive systems can lead to improved patient outcomes, enhanced efficiency, and better overall healthcare management.

Gomes et al. [33] proposed a case of integration between models of healthcare processes, created using the BPMN 2.0 standard, and the electronic medical record. Sbayou et al. [34] presents an implementation between BPMN technology and agent-based models, in order to improve performance, manage resources and ensure the coordination between workers in a healthcare context. Patient safety in modern health systems goes beyond the adoption of medical practices and standardized healthcare pathways. The increasing reliance on interconnected information systems and the generation of vast amounts of data create new challenges related to the proper handling and protection of personal data. Ensuring the privacy and security of patient information is crucial to maintaining patient trust and safeguarding their rights. In this regard, in recent years new frameworks mainly focused on ensuring patient cybersecurity have emerged, such as, for example, the CUREX conceptual model [35], which offers a platform-independent integrated environment to execute cybersecurity and risk assessments, to verify the security and robustness of information systems containing sensitive data, as well as providing a useful tool for the correct exchange of patients' information between different healthcare facilities, through the adoption of technologies such as the blockchain and IoT devices. Although the safety and privacy of patients have been strongly considered in the context of the proposed study, through the anonymization of the data provided and processed by the developed model, the study is mainly focused on the analysis and monitoring of the critical issues related to the adoption of incorrect medical practices, which could lead to serious physical harms to patients. The opportunities offered by the implementation of blockchain technology are proposed in the study conducted by Aloini et al. [36]. The adoption of blockchain and IoT technologies is also presented in Velusamy and Rajagopal [37], Du et al. [38] and Tanwar et al. [39] studies, to improve the security and reliability of healthcare systems, monitor the medical registrations on the ERP software. Based on the preliminary research conducted, it becomes evident that there is a notable absence of a real-time monitoring system for the risks that may arise during healthcare pathways within health facilities, particularly one that follows a structured and standardized approach as advocated by Object Management Group (OMG) Healthcare Domain Taskforce [40]. This particular gap in the current healthcare landscape is precisely the focus and scope of the proposed study. The lack of a comprehensive real-time monitoring system poses significant challenges for healthcare organizations in identifying and responding promptly to potential risks and adverse events. Without such a system, healthcare professionals may face difficulties in proactively managing clinical risks, resulting in potential delays in intervention and a higher likelihood of patient harm. The proposed study aims to address this gap by developing and implementing a real-time monitoring system that adheres to the structured and standardized approach proposed by OMG Healthcare Domain Taskforce [40]. This system would incorporate advanced technologies and data analytics to continuously monitor and analyze healthcare pathways, identify potential risks, and provide timely alerts to healthcare professionals. By adopting a structured and standardized approach, the study seeks to ensure that the monitoring system can be seamlessly integrated into existing healthcare processes and that its outputs are reliable and actionable. The system's implementation could potentially lead to a significant improvement in patient safety, as healthcare providers would be better equipped to identify and address risks promptly.

3. Process modeling

The objective of this study was to create a digital representation of the identified healthcare processes, through the development of some BPMN 2.0 models. The *Signavio* platform has been chosen to develop the models, in BPMN 2.0 standard notation, which provides some tools to create the executable process diagrams representations, which allows to perform some simulations on them. Through the BPMN 2.0 standard notation, it is possible to highlight all the decision-making procedures, actors, documents and information exchange involved in the different processes. The developed models can be traced back to Petri nets, in which a transition of the state associated with the system occurs upon predetermined conditions. The first version of these models had the sole objective of creating a mapping of the process. After this modeling phase, the developed models have been digitalized, to be executable, through their interfacing with the ERP software used in the analyzed healthcare facilities and the implementation of some scripts within them. In particular, the interfacing with the ERP software allows the exchange of data and information between the systems, instantiating a new process, or completing a specific task, or to exchange messages necessary for their execution, when an operator does a registration on the management software. Through this approach it was possible to keep track of the effective execution of the digitalized process diagrams, and therefore any deviation from the operating procedures, which are established by companies and based on the legislative framework. To catch the divergences from the standard procedures, the developed system returns a different status code based on the correct execution of the models. With this objective, in the preliminary study phase, the operating procedures in use at the involved healthcare facilities were analyzed. This stage of the research allowed to identify three main processes for the subsequent modeling and analysis phases:

- Laboratory analysis processes;
- Surgical room processes;
- Drug management and administration processes.

An accurate study phase for each one of the identified processes was carried out, to develop a first version of the models, as closely as possible to the operations carried out by healthcare personnel within the companies.

For the surgical room processes, the following sub-processes were identified and modeled:

- Hospitalization phase, execution of the pre-operative medical examinations and planning of the surgical intervention.
- Pre-surgical phase, concerning the preparation of the patient and his transportation and access to the surgical block, verification of the completeness of the patient's documentation.
- Surgical phase, including the records relating to the surgical and anaesthetic medical practices adopted.
- Post-surgical phase, concerning the registrations on Healthcare ERP software, such as the patient's vital parameters monitored in the phase immediately following the surgery.

The modeling of laboratory analysis processes involved the following sub-processes:

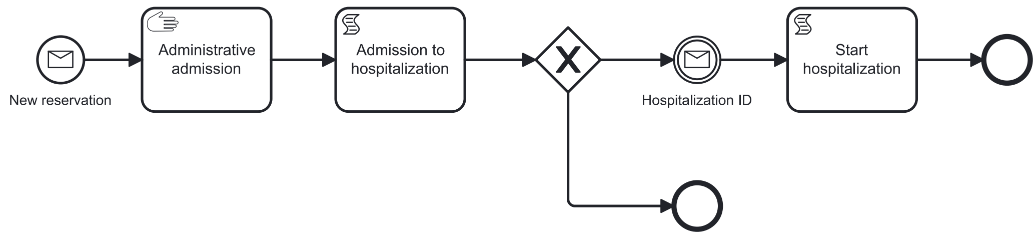

Fig. 1. Hospitalization booking sub-process.

- Pre-analytical phase, which begins with the request for laboratory analysis by the physician, continuing with the patient preparation and the collection of the blood samples by the nursing staff, and with their subsequent sorting, identified and labeled, to the analysis laboratory.
- Analytical phase, in which the actual analysis of the samples provided is performed, according to different paths for the examinations that can be performed with instrumentation interfaced with healthcare ERP software and those that can be performed manually or with non-interfaced instrumentation.
- Post-analytical phase, which concerns the communication of the results to the department and their reporting.

The models of drug management and administration processes have been divided into the following sub-processes:

- Medical examination phase, in which the patient's condition is re-evaluated, and then therapy is prescribed or updated. In the case of first access, pharmacological recognition is also carried out, as well as the patient's anamnesis, to identify the therapies already in act, allergies, and pathologies of the patient that must be taken into account for subsequent prescriptions and administrations.
- Drug administration and preparation phase, in which the nursing staff identifies the correct patient, prepares the drug and proceeds with its administration.
- Monitoring phase, in which any adverse reaction in patients due to administration is recorded.
- Pharmacological reconciliation, at the time of discharge, in which the medical staff delivers the Single Therapy Card (STC) to the patient, providing him/her with the necessary information regarding the therapy to be followed after discharge.

The adherence of the developed models (as-is configuration) to the standards identified in the preliminary study phase and to the current legislation framework was taken into account, to evaluate a possible redesign and a to-be configuration.

In the following sections, the models built for two processes are shown and some details are provided.

3.1. Patient's hospitalization process

In this section, the patient's hospitalization booking process is proposed. This process is composed of two different sub-processes, which regard respectively the patient's hospitalization booking sub-process and the proper hospitalization sub-process. The patient's hospitalization booking sub-process is shown in Fig. 1.

The circular elements represent, in the BPMN 2.0 standard, events, which can be initial, intermediate or final events. Rectangular elements, on the other hand, represent elementary tasks, although there are different types of tasks, such as script tasks, manual tasks, user tasks, etc. Some other important elements are represented by gateways, which have a rhombus shape and are used to split or join some different model branches, with an XOR or an AND logic. The hospitalization booking sub-process starts once a new booking message is received, as soon as a specially developed management software module, through a Representational State Transfer (REST) web-service, sends the Javascript Object Notation (JSON), containing the relevant booking information, as shown in Listing 1, to the proper endpoint. To select the correct endpoint the message should be sent to, the JSON message contains a *"key"* parameter, which represents the kind of task or event to trigger, while the *"businessKey"* parameter represents the unique ID assigned to the instance of the process.

```
{
 "key": "new—booking",
 "businessKey": "<booking ID>",
 "payload": {
   "initiator": "<user ID>"
 }
}
```

Listing 1: JSON request message for the registration of a new hospitalization booking instance.

Once the JSON is received by the process engine, it sends a JSON response message, shown in Listing 2.

```
{
 "key": "new—booking",
 "businessKey": "<booking ID>",
 "correlation": object,
 "payload": {
   "started": [ {
     "processId": string,
     "businessKey": "<booking ID>",
     "initiator": "<user ID>"
   } ]
 },
 "errors": [string]
}
```

Listing 2: JSON response for the registration of a new hospitalization booking instance.

The manual task, concerning *Administrative admission*, is automatically performed by the process engine, and it represents the medical procedures required at the beginning of the hospitalization period. The following script task regards the *Admission to hospitalization* task, which contains a Groovy script, executed when the task itself is instantiated. This script allows to initialize the execution variables of the *"Sanitary acceptance"*, with the value received through the JSON, while the *"initiator"*, i.e. the user who started the process instance, is assigned to the *"evaluator"* variable. The subsequent XOR gateway allows the flow to continue along only one of the outgoing branches. If the condition *$SanitaryAcceptance == "hospitalization"* is verified, meaning that the patient's hospitalization has been arranged, then the flow will continue towards the subsequent tasks, while it will be directed to an end event otherwise. If the patient is admitted to hospitalization, the flow will await the receipt of the JSON containing the user registration, in which the *"key"* field is set as *"assignment-id-shelter"*. In each one of the JSONs, there is the *"businessKey"* field, having the *booking ID* as a unique value, through which it is possible to correlate all subsequent requests to the correct process instance. When the hospitalization ID is received, the subsequent *"Start hospitalization"* script task, whose code is shown in Listing 3, is executed.

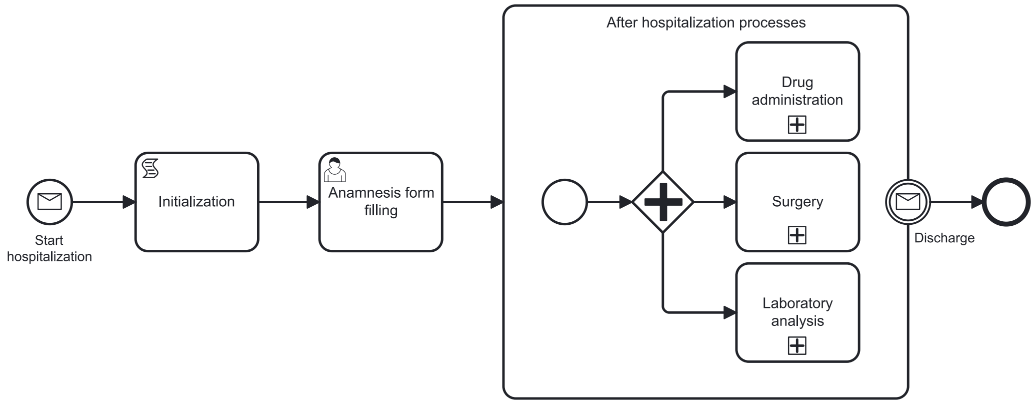

Fig. 2. Hospitalization sub-process.

```
def key = "sanitary acceptance";
def businessKey = HospitalizationIdAssignment.get("HospitalizationId");
def payload = commonService.payloadBuilder().put("initiator",execution.getVariable("evaluator")).build();
def message = messageService.createMessage(key, businessKey, null, payload);
messageService.start(message);
```

Listing 3: Groovy script for the "Start hospitalization" task.

Through the previous script, *"key"*, *"businessKey"* and *"payload"* variables are defined, within which the values of the previously illustrated process variables are entered, such as *"HospitalizationId"* and initiator. Through the method *messageService.createMessage()*, to which the previously defined variables are passed as parameters, a message is created and addressed to the proper hospitalization sub-process, shown in Fig. 2. Once the message sent by the previous sub-process is received by the second sub-process, its start event is triggered and a new instance is initialized.

The next script task allows to initialize the sub-process variables, while the next human task is performed upon receipt, from the management software module, of the respective JSON, once the anamnesis form has been filled with the patient's information. In this case, a */task/complete* REST web-service was used to interface the management software with the process engine.

```
{
    "key": "anamnesis-form-filling",
    "businessKey": "<hospitalizationID>",
    "payload": {
        "assignee": "<user ID>"
    }
}
```

Listing 4: JSON request for anamnesis form filling human task.

The patient's treatments during his hospitalization period have been modeled through the subsequent extended sub-process. The AND gateway allows the process instance to continue on the three outgoing branches, for each of which a collapsed subprocess has been inserted. Each of them refers to the relative models created for the processes of drug administration, surgery and laboratory analysis. When the process instance reaches one of these sub-processes, it is instantiated and executed. The *Drug administration* process is explained, as an example, in Section 3.2. The boundary event of receipt of the patient's discharge communication message is instantiated upon registration on the management software, which corresponds to the sending of a JSON containing the *"key":"Discharge"*, and with a structure like those previously illustrated. Upon receipt of the discharge message, the corresponding process instance is closed.

3.2. Drugs administration process

In this section, the workflow concerning the drug administration to the patient, shown in Fig. 3, is described.

In this case, the process flow is instantiated once the previous process instance reaches the *Drug administration* sub-process, as shown in the previous section. After the start event, a script task is executed, to create the patient's drug list. This script task sets the execution variables shown in Listing 5, which represent the patient's drugs, the new prescriptions and the suspended ones.

```
execution.setVariable("patientMedication", []);
execution.setVariable("prescriptions", [:]);
execution.setVariable("cancelledPrescriptions", []);
```

Listing 5: Execution variable to be set once the process is instantiated.

The subsequent sub-process contains some user tasks, such as drug reconciliation, taking charge of the patient's drug, prescription of the therapy and its validation, connected through an AND gateway. Due to this type of connection, a single instance could run through each branch of the process flow, and the user tasks could be performed simultaneously. The execution of each one of the tasks is related to the exchange of some information between the process models and the ERP software, through different specially created web-services. An endpoint, of type */task/complete*, is linked to each one of the user tasks, and they are completed once the proper JSON request message is received and the JSON response is sent back to the ERP software. The JSON request message sent from the ERP software to the process module for the therapy prescription is shown, as an example, in Listing 6.

```
{
    "key": "therapy-prescription",
    "businessKey": "<IDhospitalization>",
    "payload": {
        "assignee": "<IDdoctor>",
        "outcomes": {
            "prescription": {
                "id": "<IDprescription>",
                "code": "<IDmedication>",
                "warehouse": "<IDwarehouse>",
                "depository": "<IDdepository>",
                "date": {
                    "from": "<YYYY-MM-DD>",
                    "to": "<YYYY-MM-DD>",
                    "hours": [{
                        "hour": "<hh:mm>",
                        "qt":"<quantity>"
                    },
                    {
```

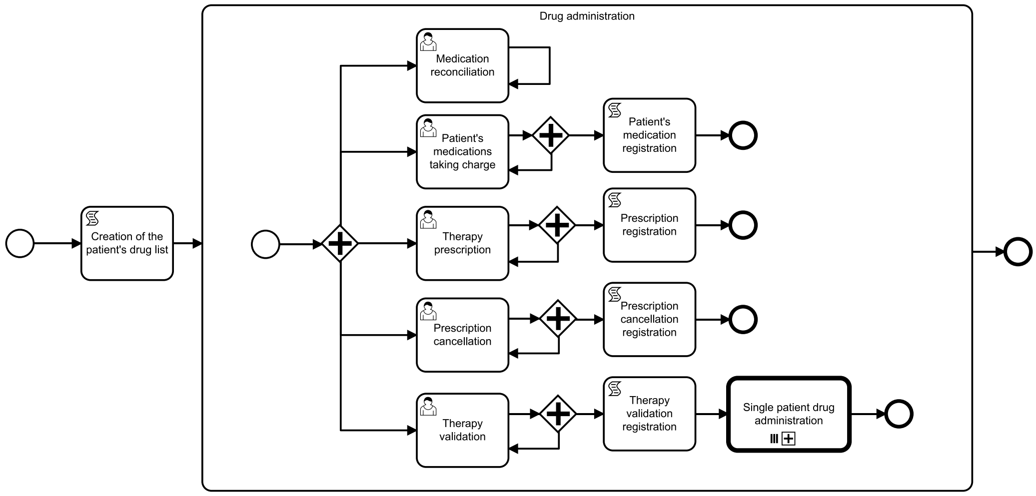

Fig. 3. Drugs administration process.

```
                                ...
                            }
                        ]
                    }
                }
            }
        }
}
```

Listing 6: JSON request message for the Therapy prescription task.

The JSON response message is shown in Listing 7.

```
{
    "key": "therapy—prescription",
    "businessKey": "<IDhospitalization>",
    "correlation": object,
    "payload": {
                "completed": [{
                    "processId": string,
                    "businessKey": "<
                        IDhospitalization>",
                    "taskId": string,
                    "assignee": "<IDdoctor>"
                }]
            },
    "errors": [string]
}
```

Listing 7: JSON response message for the Therapy prescription task.

The execution of multiple drug prescriptions, suspensions or validations, was modeled through the subsequent AND gateways, which allow to execute both the next script tasks for the single drug and the previous user tasks for a new one. As an example, the Groovy code contained in the *"Prescription registration"* script task is shown in Listing 8.

```
def prescription = execution.getVariable("
     prescription");
prescription.validated = false;
def prescriptions = execution.getVariable("
     prescriptions");
presriptions.put(prescription.id, prescription);
execution.setVariable("prescriptions",
     prescriptions);
```

Listing 8: Groovy code in Prescription registration script task.

On the last process branch, which regards the validation of the therapy, the *"Single patient drug administration"* task is instantiated. This sub-task is executed in a parallel multi-instance loop, as multiple drugs, registered in the previous script tasks, have to be administered to the same patient.

4. Risk analysis

After completing the modeling phase, a comprehensive risk analysis was carried out using the Healthcare Failure Mode and Effect Analysis (H-FMEA) methodology, which is the most common risk assessment tool used for patient safety [41–46]. This approach allowed for a thorough examination of each step in the healthcare pathways, identifying potential failure modes, their effects, and the corresponding risk levels. The risk analysis aimed to understand the critical points in the processes and implement appropriate preventive measures to ensure patient safety and quality of care. During the risk analysis, the following steps were undertaken:

- Identification of Risks: Each step in the healthcare pathways was analyzed to identify potential failure modes or events that could lead to adverse outcomes. These risks were documented and categorized based on their severity and likelihood of occurrence.
- Calculation of Risk Priority Number (RPN): The RPN was determined for each identified risk. This number is calculated by multiplying three factors: Severity of the injury or damage that the patient may suffer (S), Likelihood or probability that the event happens (O), and capability of Detection (D) i.e. the ease or difficulty of detecting the error before it causes damage. The RPN helps prioritize the risks, with higher values indicating a greater need for immediate attention and mitigation. The RPN also allows to build statistics on the phases most affected by errors within the same process, allowing the identification of causes and containment factors.
- Identification of Consequential Damages: For each risk, the potential effects or damages to patients and the healthcare system were assessed. Understanding the consequences of each risk allowed for better decision-making and resource allocation to prevent or mitigate these effects.
- Adoption of Prevention Measures: Based on the risk analysis results, appropriate prevention measures were identified and implemented. These measures were designed to reduce the likelihood of occurrence and/or severity of the identified risks. They included process improvements, changes in protocols, training for healthcare staff, and the integration of technological solutions.

Table 1
Subset of key performance indicators implemented in the proposed system.

	Indicator	Value
(1)	Postponed surgeries due to incomplete documentation	0%
(2)	Validated drugs prescriptions against drugs prescribed	98%
(3)	Adverse reactions with respect to administrations	0%
(4)	Average length of stay	4.30 days
(5)	Mean medical assistance time per patient	309,48 min/day
(6)	Mean time between sanitary acceptance and hospitalization	6,25 days
(7)	Mean time between hospitalization and surgery	1,04 days
(8)	Turnover rate	19,14

Following the outcomes of the H-FMEA analysis and considering the relevant Ministerial guidelines, a comprehensive set of indicators was developed. These indicators were specifically tailored to monitor the processes modeled in the previous phases and were seamlessly integrated with the existing indicators already employed in healthcare structures. The primary objective of this indicator set is to continuously monitor the advancement of the processes and ensure the accurate execution of each phase, thus preventing any errors resulting from deviations from established company procedures. To achieve this, the indicators were designed to be measured over various time frames, allowing for a holistic view of the process performance and enabling timely intervention if any anomalies or risks are detected. The indicator set encompasses a variety of key performance metrics that reflect different aspects of the healthcare pathways. Some of the indicators focus on short-term monitoring, enabling real-time assessment of critical activities and rapid response to potential issues, thus representing a tool capable of reducing the incidence or severity of the risk factors found within the processes in the previous stages. Others target medium and long-term evaluation, offering insights into the overall efficiency and effectiveness of the processes and the adherence to established standards over extended periods. The selected indicators encompass both process-related metrics and outcome-based measurements. Process-related indicators gauge the execution of each step in the healthcare pathways, ensuring compliance with the prescribed procedures and identifying any deviations early on. Outcome-based indicators, on the other hand, measure the actual results and patient outcomes, providing valuable insights into the overall quality and safety of care delivered. As an example, some of the indicators calculated thanks to the implementation of the proposed solution are shown in Table 1. The indicators with index (1), (2) and (3) are examples of process-related ones, as they increase with processes non-conformities. Those with indexes (4), (5), (6), (7) and (8) are outcome-based indicators, as they highlight process performances over a medium-term time interval.

This set of indicators was used to create a new software module which allows the clinics to extract and analyze data from the respective databases, providing clinics with valuable insights into their performance and patient outcomes. By deploying this indicator set, the healthcare structures can proactively identify potential risks, inefficiencies, or deviations in their healthcare processes and take appropriate corrective actions promptly. Continuous monitoring of the indicators can contribute to a data-driven approach to decision-making, enabling healthcare professionals to make informed choices to optimize patient care and improve overall healthcare performance. The periodic evaluation of the indicator data will facilitate trend analysis, benchmarking against predefined targets, and the identification of improvement opportunities. Regular feedback and reporting on the indicator set will foster a culture of continuous quality improvement within the healthcare structures, enhancing patient safety and the overall quality of healthcare services. The user-friendly interface of the software module makes it accessible to a wide range of healthcare professionals, from administrators to clinicians, enhancing decision-making and facilitating collaboration among different stakeholders.

5. System integration

Three different types of REST web services were used to interface the models created in the project with the management software used in the clinics (as shown in Fig. 4):

- *Message/start*, when received by the engine, corresponds to the start of an instance of a specific process.
- *Message/send*, through which data is exchanged between the engine and the management software.
- *Task/complete*, the receipt of which by the engine corresponds to the completion of a specific instance of a thread.

The seamless exchange of information between the process engine, responsible for executing the developed process models, and the Healthcare ERP software, through which operators interact, enables the completion of specific tasks (elementary actions) within the created models. This dynamic facilitates continuous monitoring of task statuses in real-time by triggering relevant tasks and instances through the integrated layer of Web Services in the Process Engine. The implementation and interfacing with the process models have been thoughtfully designed to operate discreetly in the background, ensuring minimal disruption to the regular operations of medical and nursing staff. The system has been meticulously engineered to avoid generating alerts or error messages, preserving the smooth functioning of the clinics' software. These comprehensive implementations involved not only the development, digitalization, and execution of process models but also necessitated the revision of software used in the respective facilities to enable seamless interfacing with the cloud-based process engine. Through the coordination with the process engine, based on the messages exchanged with the ERP software, it was possible to collect information about the most critical processes concerning the operating procedures established by the management of the structures. Additionally, the system adeptly tracks major process non-conformities, promptly identifying instances where tasks deviate from the models, thereby potentially posing risks to patients during their clinic stay. To ensure comprehensive record-keeping and analysis, all execution information has been systematically stored in a dedicated process execution database. During the subsequent testing phase, the primary objectives were to verify the system's flawless functioning within the implemented workflows, seamless interfacing with existing systems, and to ascertain its long-term stability and maintenance capability, especially concerning data transfer overload and communication exchanges. Upon successfully validating the system's accuracy, functionality, and stability, the test environment was meticulously replicated on the servers of both clinics. This replication involved updating the pre-existing software versions and installing and configuring a dedicated server exclusively for the Business Process Management Suite (BPMS) system. This sophisticated and carefully orchestrated setup ensures that the updated system runs efficiently and consistently within the clinics' operational infrastructure.

6. Experimental phase and results

During the final project phase, the focus shifted to the real-world implementation and utilization of the new system, which was now interfaced with the process engine. The primary objective was to gather data on how the system functioned in practical scenarios within the healthcare facilities. This phase was crucial in evaluating the system's effectiveness in supporting and enhancing the healthcare processes. An essential aspect of this phase was the training of healthcare workers on how to use the new system effectively. Proper training ensured

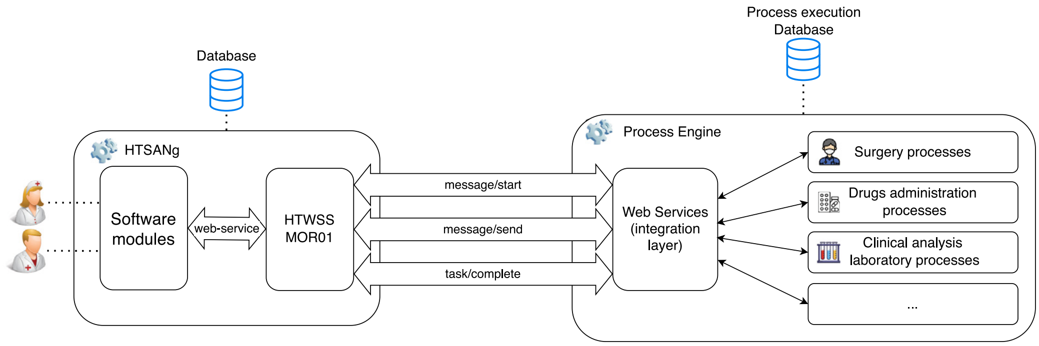

Fig. 4. System integration.

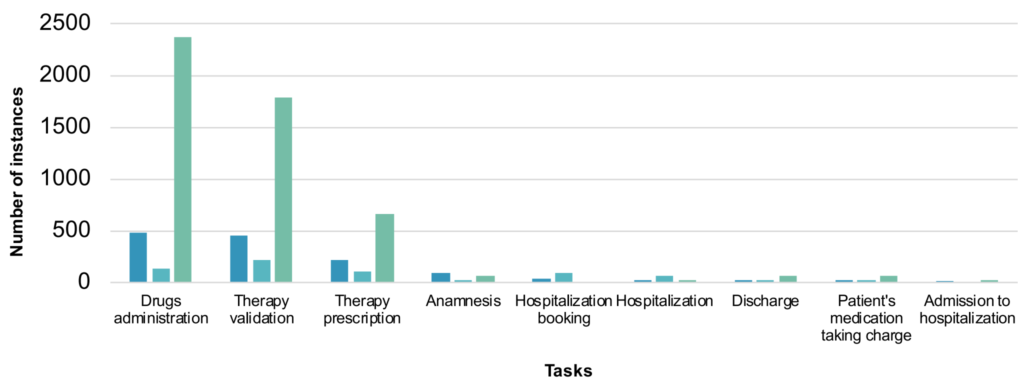

Fig. 5. Status code returned in different tasks in the Drug Administration process.

that the healthcare professionals understood the functionalities and capabilities of the system and could utilize it to its full potential. The experimental phase was conducted on the job, through normal registration operations within the ERP software by healthcare workers involved. For this phase, no constraints or alerts were inserted in the system. This phase allowed the identification of bottlenecks within the processes and non-conformities in their execution, through the analysis of the status codes returned by the system. As regards the status codes for the process instances, code 0 represents a communication error between the software and the BPMS server, code 1 represents the correct execution of the single process instance, and code 100 represents discrepancies concerning the company's operating policies. Throughout this phase, data was collected from the system, capturing key performance indicators and user feedback. The data allowed the project team to identify any potential issues or areas for improvement in the system's functionality and user experience. By analyzing this data, the team could fine-tune the system and address any discrepancies or inefficiencies. The project team also engaged in ongoing communication with healthcare professionals to gather their insights and opinions on the system's usability and impact on their daily tasks. This feedback played a vital role in refining the system's design and functionality based on real-world user experiences. The distribution of each status code is represented in the histogram shown in Fig. 5, which groups them by single task in the drug administration process.

The histogram in Fig. 6, on the other hand, shows the distribution of status code 100, which therefore represents an operational non-conformity, concerning the different tasks of the drug administration process, taken as an example. This graph made it possible to focus attention on the sub-processes most affected by procedural errors, therefore potentially having the greatest impact on the success of the care pathways and the safety of the patient during the hospitalization period.

By the end of this phase, the project team had a comprehensive understanding of the system's performance and its value in optimizing healthcare operations. The successful implementation of the system, coupled with positive user feedback, confirmed its potential to change healthcare practices, enhance patient care, and strengthen compliance with regulatory standards.

7. Conclusions

The use of a structured approach in a context dominated by human work, where healthcare workers are responsible for critical decisions, has effectively revealed discrepancies and anomalies that may arise in daily activities. The adoption of BPM technology, specifically the BPMN 2.0 notation, has proven to be a wise choice to ensure process compliance with applicable regulations and international standards.

The integrated system developed has demonstrated its ability to monitor deviations from established operating procedures at the organizational level, which are based on ministerial standards and recommendations. Such deviations could pose risks to patients, with potentially serious reputational and financial consequences for the healthcare organization, as well as physical harm to the patient. The introduction of Key Performance Indicators (KPIs) provides valuable insight for medium to long-term monitoring, evaluating improvements following the implementation of restrictions and alerts in the management software. This immediate reporting of operational discrepancies to operators increases the effectiveness of the system.

Future development of the proposed work involves the incorporation of constraints and alerts within the ERP software. This enhancement would be based on the background execution of the developed process models, aiming to restrict the operator from undertaking specific tasks if they are executed out of the expected order or deviate from the prescribed operating procedures. By implementing such constraints, the system can ensure adherence to the defined processes, enhancing process consistency and minimizing the potential for errors.

Another area of potential advancement pertains to the modeling of specific aspects of medical procedures characterized by their unstructured nature, primarily comprising decision-making processes. To address these scenarios, we will explore the adoption of a model

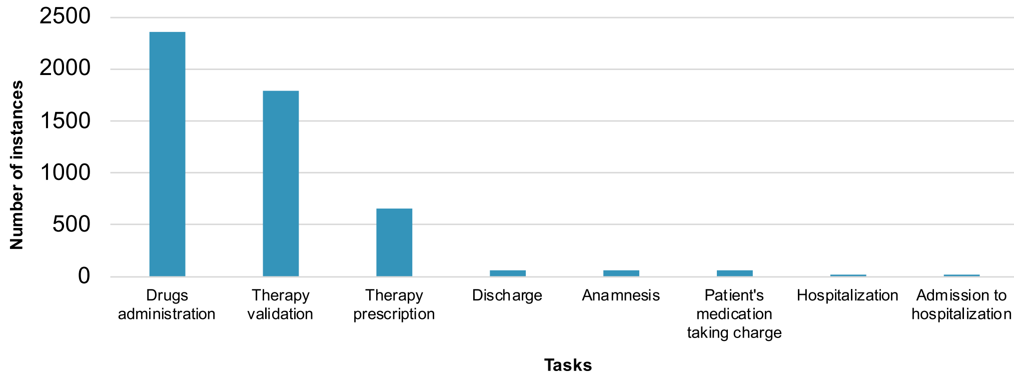

Fig. 6. Status code "100" distribution.

developed using the Case Management Model and Notation (CMMN) standard for unstructured processes or the Decision Model and Notation (DMN) for decision processes. Such modeling approaches enable a more comprehensive representation of complex decision processes, ensuring clarity and efficiency in handling critical medical decisions.

Through these proposed developments, the study seeks to further optimize the healthcare processes, streamline operational workflows, and enhance the overall quality and safety of patient care.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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